

IT PROFESSIONAL PORTFOLIO

Cloud Engineering and Security Projects

PROJECT 13

Enable IAM Identity Center, Enforce MFA, and Create SSO User and Admin Group

IAM Identity Center | SSO | MFA | Group-Based Access

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ENVIRONMENT	Personal AWS Lab
VERSION	v1.0.0
RELEASE DATE	July 13, 2026
STATUS	Complete

Executive Summary

Enabled an organization instance of **AWS IAM Identity Center**, configured the Identity Center directory as the identity source, enforced MFA for AWS access portal sign-ins, created an **Administrators** group, created the **Msalazar** SSO user account, and added that user to the Administrators group. This project establishes the portfolio's workforce identity and SSO foundation.

Business Problem

Managing access with separate IAM users in every AWS account does not scale and increases credential risk. IAM Identity Center provides a central AWS access portal, a workforce identity store, group-based access management, and a path toward federated access using enterprise identity providers. Requiring MFA for Identity Center sign-ins strengthens authentication before users can access assigned AWS accounts or applications.

Objectives

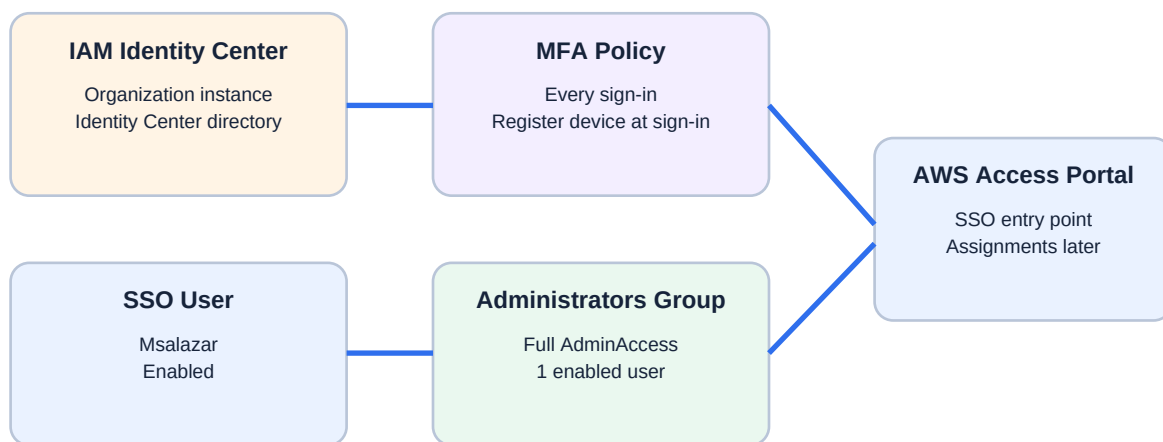
- Enable IAM Identity Center for the AWS organization.
- Use the Identity Center directory as the identity source.
- Configure MFA to prompt users every time they sign in.
- Require users without a registered MFA device to register one at sign-in.
- Create an Administrators group for future AWS account assignments.
- Create the Msalazar SSO user account.
- Add Msalazar to the Administrators group.
- Document that account permission assignments are not yet included in this lab.

Environment

Cloud Provider	Amazon Web Services (AWS)
Primary Service	IAM Identity Center
Identity Source	Identity Center directory
SSO User	Msalazar / Michael Salazar
User Status	Enabled
Group	Administrators
Group Description	Full AdminAccess
MFA Prompt	Every time users sign in

MFA Registration	Require registration at sign-in if no MFA device exists
Allowed MFA Types	Authenticator apps, security keys, and built-in authenticators
Evidence Limitation	User list shows MFA devices as None at capture time; this proves registration had not yet occurred, while the policy requires registration at sign-in.
Future Step	Permission sets and AWS account assignments

Architecture



Identity Center centralizes workforce sign-in, groups, and MFA requirements for AWS access portal sessions.
Account permission-set assignments are intentionally documented as next-step work, not completed here.

Figure 1 - IAM Identity Center SSO user, group, and MFA foundation.

Implementation Summary

1. Opened IAM Identity Center from the AWS management account.
2. Created the organization instance of IAM Identity Center.
3. Confirmed the Identity Center directory as the identity source.
4. Reviewed the AWS access portal URL and redacted sensitive portal details for the portfolio evidence.
5. Configured MFA to prompt users every time they sign in.
6. Required users without a registered MFA device to register an MFA device at sign-in.
7. Created an Administrators group with a Full AdminAccess description.
8. Created the Msalazar SSO user account with Enabled status.

9. Added Msalazar to the Administrators group.

SCOPE

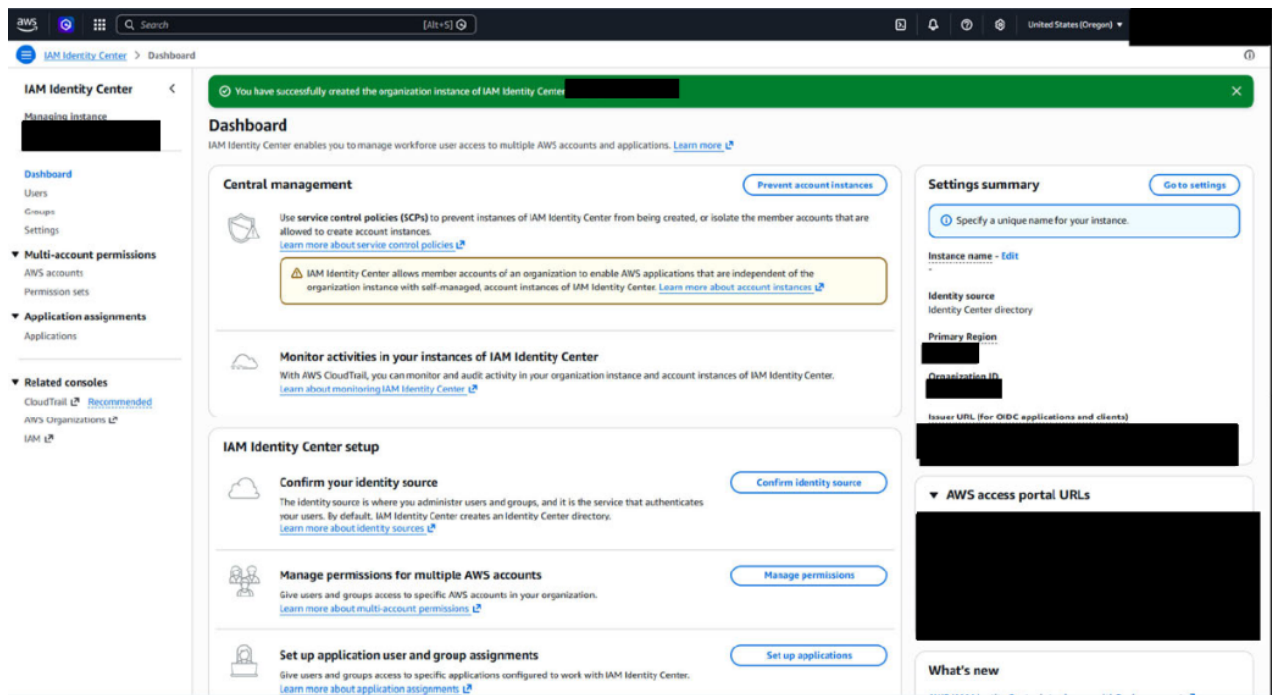
This project creates the **identity foundation**. It does not yet prove account-level authorization because permission sets and AWS account assignments are a separate follow-up step.

Federation and SSO Concepts

Identity Provider	IAM Identity Center directory authenticates the workforce user in this lab.
Relying Party	AWS accounts and applications rely on Identity Center for sign-in and session identity.
Single Sign-On	Users sign in through the AWS access portal instead of maintaining separate IAM users per account.
Group-Based Access	Administrators group will be used later to assign permission sets to AWS accounts.
MFA Requirement	Every Identity Center access portal sign-in requires MFA behavior before account access.

Evidence

Figure 2 - IAM Identity Center Organization Instance Created



The IAM Identity Center dashboard shows a success banner confirming the organization instance was created. The Settings summary shows the Identity Center directory as the identity source. Sensitive portal and issuer values were redacted before publication.

Figure 3 - MFA Required for Identity Center Sign-In

Multi-factor authentication Configure

Define the behavior that you want to enforce to secure access to the AWS access portal with multi-factor authentication (MFA). [Learn more](#)

Prompt users for MFA
Every time they sign in (always-on)

When prompted for MFA
Users can authenticate with: Authenticator apps, security keys, and built-in authenticators

If this user does not have a registered MFA device
Require them to register an MFA device at sign in

Who can manage MFA devices
Users and administrators can add and manage MFA devices

The Multi-factor authentication settings show that users are prompted for MFA every time they sign in, that users without a registered MFA device must register one at sign-in, and that authenticator apps, security keys, and built-in authenticators are allowed.

Figure 4 - SSO User Created and Enabled

Users (1) Delete users Add user

Users listed here can sign in to the AWS access portal to access AWS accounts and assigned cloud applications. [Learn more](#)

Username

☐	Username	Display name	Status	MFA devices	Created by
☐	Msalazar	Michael Salazar	Enabled	None	Manual

The Identity Center Users page shows one user: Msalazar, display name Michael Salazar, status Enabled, created manually. The MFA devices column shows None at the time of evidence capture, which is why registration at sign-in remains important.

Figure 5 - Administrators Group with User Membership

Administrators Delete group

General Information Edit description

Group name Administrators	Created time July 13, 2026	Last updated July 13, 2026
	Created by Manual	Updated by Manual
Description Full AdminAccess		

Users **AWS accounts** **Applications**

Users in this group (1) Remove users from group Add users to group

Workforce users in this group inherit permissions to the AWS accounts and Identity Center enabled applications that are assigned to this group.

Find users by username or display name

☐	Username	Display name	Status	Email
☐	Msalazar	Michael Salazar	Enabled	

The Administrators group page shows the group name Administrators, description Full AdminAccess, creation and update date of July 13, 2026, and one enabled user, Msalazar, in the group. Sensitive email and identifier values were redacted before publication.

Security Analysis

IAM Identity Center reduces reliance on long-lived IAM users in each AWS account by centralizing workforce sign-in through the AWS access portal.

The MFA configuration is stronger than relying only on email-based first-device verification because the access portal is configured to prompt for MFA every sign-in.

Group-based administration is more scalable than assigning access directly to individual users. The Administrators group can later be assigned permission sets across selected AWS accounts.

The evidence shows that the user did not yet have an MFA device registered at capture time. The configured policy compensates by requiring registration at sign-in before normal use.

The lab guide highlights that Identity Center can use its built-in user pool as the identity provider or bridge to external identity providers such as enterprise directories, which is important for future production maturity.

Lessons Learned

Federation separates authentication from authorization. The identity provider proves who the user is, while AWS account assignments and permission sets decide what the user can do.

SSO is a foundation for scalable cloud access because users can authenticate through a central portal rather than managing separate IAM credentials in every account.

Groups should be created before broad account assignments so access can be managed by role or job function rather than by individual exception.

MFA policy and MFA enrollment are related but distinct. A policy can require MFA registration even if the user has not yet completed registration.

Production Improvements

- Create permission sets and assign the Administrators group only to accounts that require administrative access.
- Use least-privilege permission sets for routine work instead of granting full administrative access everywhere.
- Integrate IAM Identity Center with an external identity provider such as Microsoft Entra ID, Okta, or another enterprise IdP when available.
- Use an SCP to prevent unapproved account-level Identity Center instances.
- Require phishing-resistant MFA such as security keys where possible for privileged groups.
- Monitor Identity Center user creation, group changes, MFA changes, and permission assignments through CloudTrail and centralized logging.

Skills Demonstrated

IAM Identity Center	MFA enforcement	AWS Organizations
SSO user creation	Group-based access	Identity Center directory
Federation concepts	Privileged group setup	Access portal foundation

Resume Bullet

RESUME

Configured AWS IAM Identity Center as the organization's SSO foundation by enabling the organization instance, enforcing MFA for every access portal sign-in, creating an Administrators group, and adding an enabled SSO user for future permission-set assignments.

STAR Interview Story

Situation: The AWS portfolio was using account-based administrative patterns and needed a scalable workforce identity foundation for multi-account access.

Task: Enable IAM Identity Center, enforce stronger authentication, and create the first SSO user and administrative group.

Action: I enabled the IAM Identity Center organization instance, configured always-on MFA requirements, created the Administrators group, created the Msalazar user, and added the user to the group.

Result: The environment now has a centralized SSO and MFA foundation ready for future permission-set and AWS account assignments.

Version History

v1.0.0	July 13, 2026	Initial documented implementation. Enabled IAM Identity Center, configured always-on MFA requirements, created the Administrators group, created the Msalazar SSO user, added the user to the group, documented MFA enrollment and account-assignment limitations, and embedded all provided project evidence using portfolio documentation standard v1.0.1.

References

- Cloud Security Lab a Week (S.L.A.W.) - Bring in the Fed(eration): SSO.
- User-provided IAM Identity Center enabled screenshot.
- User-provided MFA configuration screenshot.
- User-provided Identity Center user list screenshot.
- User-provided Administrators group membership screenshot.
- Project 12 - Organization CloudTrail Centralized Logging.